GAOPEI PAN

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RESEARCH/WORK EXPERIENCE

University of Würzburg, Germany *May.2024 – Present*
PostDoc in Physics, Supervisor: Prof. Fakher Assaad
The University of Hong Kong, China *Sept.2023 – Jan.2024*
Research Assistant in Physics (waiting for visa of Germany)

EDUCATION

Institute of Physics, Chinese Academy of Sciences, China *Sept.2018 – Jul.2023*
Ph.D. in Physics, Supervisor: Prof. Zi Yang Meng
University of Chinese Academy of Sciences, China *Sept.2014– Jul.2018*
B.S. in Physics Major with Computer Science Minor

RESEARCH SUMMARY

My research focuses on quantum many-body physics, including non-Fermi-liquid behavior, quantum criticality, correlated phases, and superconductivity in strongly interacting fermion systems and moiré materials. I also work on entanglement-entropy–related problems and am currently exploring gauge-field–coupled Kondo systems. In addition to model studies, I specialize in numerical method development, particularly Quantum Monte Carlo techniques such as determinantal QMC, momentum-space QMC, entanglement-entropy algorithms, self-learning Monte Carlo, and the sign problem in QMC.

(See research statement for details)

SELECTED PUBLICATIONS

1. Quantum Monte Carlo studies of U(1) lattice gauge models of Kondo breakdown
 Gaopei Pan, Fakher F. Assaad
 arXiv:2512.17801

2. High-efficiency quantum Monte Carlo algorithm for extracting entanglement entropy in interacting fermion systems
 Weilun Jiang, **Gaopei Pan**[†], Zhe Wang, Bin-Bin Mao, Heng Shen, Zheng Yan
 arXiv:2409.20009

3. Integral algorithm of exponential observables for interacting fermions in quantum Monte Carlo simulations
 Xu Zhang, **Gaopei Pan**, Bin-Bin Chen, Kai Sun, Zi Yang Meng
 Phys. Rev. B 110, L121112 (2024)

4. Stable computation of entanglement entropy for two-dimensional interacting fermion systems
 Gaopei Pan, Yuan Da Liao, Weilun Jiang, Jonathan D'Emidio, Yang Qi, Zi Yang Meng
 Phys. Rev. B 108, L081123(2023)

5. Thermodynamic characteristic for correlated flat-band system with quantum anomalous Hall ground state
 Gaopei Pan, Xu Zhang, Hongyu Lu, Heqiu Li, Bin-Bin Chen, Kai Sun, Zi Yang Meng
 Phys. Rev. Lett. 130, 016401 (2023)

PUBLICATIONS

1. Quantum Monte Carlo studies of U(1) lattice gauge models of Kondo breakdown
Gaopei Pan, Fakher F. Assaad
arXiv:2512.17801

2. Self-learning Monte Carlo Method: A Review
Gaopei Pan, Chuang Chen, Zi Yang Meng
arXiv:2507.12554 (Book chapter)

3. Loop-Current Fluctuations Mediated Chiral d-Wave Pairing in Kagome Latticen
Qi-Fang Li, **Gaopei Pan**, Xu Zhang, Satoru Nakatsuji, Weilun Jiang, Xiao Yan Xu and Xianxin Wu
Chin. Phys. Lett. 42 057302 (2025)

4. Quasiparticle Spectroscopy of Chiral Charge Order
Jiangchang Zheng*, Caiyun Chen*, **Gaopei Pan***, Xu Zhang*, Chen Chen, Yuan Da Liao, Ganesh Pokharel, Andrea Capa Salinas, Yizhou Wei, Hoi Chun Po, Ding Pan, Stephen D. Wilson, Zi Yang Meng, Berthold Jäck
arXiv:2503.19032

5. Reweight-annealing method for calculating the value of partition function via quantum Monte Carlo
Yi-Ming Ding, Jun-Song Sun, Nvsen Ma, **Gaopei Pan** , Chen Cheng, Zheng Yan
Phys. Rev. B 110, L121112 (2024)
arXiv:2403.08642

6. High-efficiency quantum Monte Carlo algorithm for extracting entanglement entropy in interacting fermion systems
Weilun Jiang, **Gaopei Pan**†, Zhe Wang, Bin-Bin Mao, Heng Shen, Zheng Yan
arXiv:2409.20009

7. Defining a universal sign to strictly probe a phase transition
Nvsen Ma, Jun-Song Sun, **Gaopei Pan**†, Chen Cheng, and Zheng Yan
Phys. Rev. B 110, 125141 (2024)
arXiv:2301.12438

8. Universal collective Larmor-Silin mode emerging in magnetized correlated Dirac fermions
Chuang Chen, Yuan Da Liao, Chengkang Zhou, **Gaopei Pan**, Zi Yang Meng, Yang Qi
Phys. Rev. B 110, L121112 (2024)
arXiv:2401.14358

9. Integral algorithm of exponential observables for interacting fermions in quantum Monte Carlo simulations
Xu Zhang, **Gaopei Pan**, Bin-Bin Chen, Kai Sun, Zi Yang Meng
Phys. Rev. B 110, L121112 (2024)
arXiv:2311.03448

10. Disorder Operator and Rényi Entanglement Entropy of Symmetric Mass Generation
Zi Hong Liu, Yuan Da Liao, **Gaopei Pan**, Weilun Jiang, Chao-Ming Jian, Yi-Zhuang You, FakherF Assaad, Zi Yang Meng, Cenke Xu
Phys. Rev. Lett. 132, 156503 (2024)
arXiv:2308.07380

11. Evolution from quantum anomalous Hall insulator to heavy-fermion semimetal in twisted bilayer graphene
Cheng Huang, Xu Zhang, **Gaopei Pan**, Heqiu Li, Kai Sun, Xi Dai, Zi Yang Meng

Phys. Rev. B 109, 125404 (2024)
arXiv:2304.14064

12. Stable computation of entanglement entropy for two-dimensional interacting fermion systems
Gaopei Pan, Yuan Da Liao, Weilun Jiang, Jonathan D'Emidio, Yang Qi, Zi Yang Meng
Phys. Rev. B 108, L081123(2023)
arXiv:2303.14326

13. Polynomial sign problem and topological Mott insulator in twisted bilayer graphene
Xu Zhang, **Gaopei Pan**, Bin-Bin Chen, Heqiu Li, Kai Sun, Zi Yang Meng
Phys. Rev. B 107, L241105(2023)
arXiv:2210.11733

14. The teaching from entanglement: 2D deconfined quantum critical points are not conformal
Yuan Da Liao, **Gaopei Pan**, Weilun Jiang, Yang Qi, Zi Yang Meng
arXiv:2302.11742

15. Thermodynamic characteristic for correlated flat-band system with quantum anomalous Hall ground state
Gaopei Pan, Xu Zhang, Hongyu Lu, Heqiu Li, Bin-Bin Chen, Kai Sun, Zi Yang Meng
Phys. Rev. Lett. 130, 016401 (2023)
arXiv:2207.07133

16. A Sport and a Pastime: Model Design and Computation in Quantum Many-Body Systems
Gaopei Pan, Weilun Jiang, Zi Yang Meng
Chin. Phys. B 31 127101 (2022)

17. Superconductivity and bosonic fluid emerging from Moiré flat bands
Xu Zhang, Kai Sun, Heqiu Li, **Gaopei Pan**, Zi Yang Meng
Phys. Rev. B 106, 184517 (2022) Editors' Suggestion
arXiv:2111.10018

18. Superconductivity near the $(2+1)$d ferromagnetic quantum critical point
Yunchao Hao*, **Gaopei Pan***, Kai Sun, Zi Yang Meng, Yang Qi
Chin. Phys. Lett. 39 097102 (2022)

19. Fermion sign bounds theory in quantum Monte Carlo simulation
Xu Zhang, **Gaopei Pan**, Xiao Yan Xu, Zi Yang Meng
Phys. Rev. B 106, 035121 (2022)
arXiv:2112.06139

20. Sign Problem in Quantum Monte Carlo Simulation
Gaopei Pan, Zi Yang Meng
arXiv:2204.08777 (Book chapter for Elsevier Encyclopedia of Condensed Matter Physics).

21. Solving quantum rotor model with different Monte Carlo techniques
Weilun Jiang, **Gaopei Pan**, Yuzhi Liu, Zi Yang Meng
Chin. Phys. B 31, 040504 (2022)
arXiv:1912.08229

22. Dynamical properties of collective excitations in twisted bilayer Graphene
Gaopei Pan, Xu Zhang, Heqiu Li, Kai Sun, Zi Yang Meng
Phys. Rev. B 105, L121110 (2022)
arXiv:2102.10755

23. Momentum space quantum Monte Carlo on twisted bilayer Graphene
Xu Zhang*, **Gaopei Pan***, Yi Zhang, Jian Kang, Zi Yang Meng

Chin. Phys. Lett. 38, 077305 (2021) Cover story
arXiv:2105.07010

24. Phase diagram of the spin-1/2 Yukawa-SYK model: Non-Fermi liquid, insulator, and superconductor
Wei Wang, Andrew Davis, **Gaopei Pan**, Yuxuan Wang, Zi Yang Meng
Phys. Rev. B 103, 195108 (2021)
arXiv:2102.10755

25. Yukawa-SYK model and Self-tuned Quantum Criticality
Gaopei Pan, Wei Wang, Andrew Davis, Yuxuan Wang, Zi Yang Meng
Phys. Rev. Research 3, 013250 (2021)
arXiv:2001.06586

26. Itinerant Quantum Critical Point with Fermion Pockets and Hot Spots
Zi Hong Liu, **Gaopei Pan**, Xiao Yan Xu, Kai Sun, Zi Yang Meng
PNAS August 20, 2019 116 (34) 16760-16767
arXiv:1808.08878

27. Revealing Fermionic Quantum Criticality from New Monte Carlo Techniques
Xiao Yan Xu, Zi Hong Liu, **Gaopei Pan**, Yang Qi, Kai Sun, Zi Yang Meng
TOPICAL REVIEW, J. Phys.: Condens. Matter 31, 463001 (2019)
arXiv:1904.07355